Assignment-3 Combinational Assignment

1) Implement the following functions using (i) 2:1 MUX and NOT gates (ii) 4:1 MUX

(a) $F(A,B,C) = \overline{A}C + B\overline{C}$

→ (i) using 2:1 MUX & NOT gate.

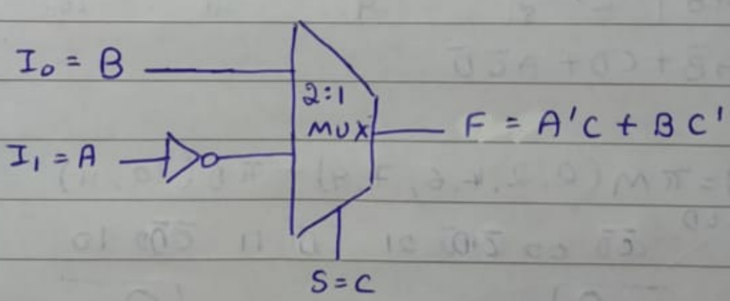
using select line, $S = C$

- when $C=0$ → $I_0 = B$
- when $C=1$ → $I_1 = A'$

$\therefore F = C' I_0 + C I_1 = C'(B) + C(A')$

$F = A'C + BC'$

| A | B | C | A' | C' | F = A'C + BC' |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 | 0 |



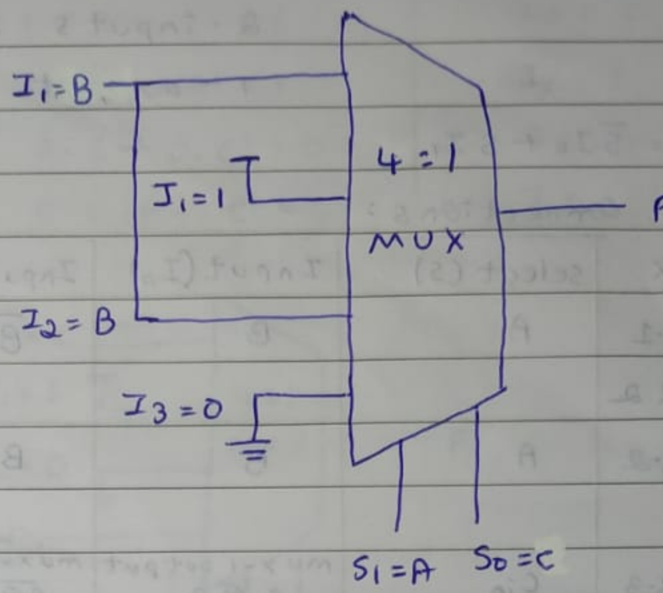

(ii) using 4:1 MUX :

using select lines, $S_1 = A$ $S_0 = C$

The mux inputs are:

| A | C | $F = \overline{A}C + B\overline{C}$ | mux input |
|---|---|---|---|
| 0 | 0 | 0 + B = B | $I_0 = B$ |
| 0 | 1 | 1 + 0 = 1 | $I_1 = 1$ |
| 1 | 0 | 0 + B = B | $I_2 = B$ |
| 1 | 1 | 0 + 0 = 0 | $I_3 = 0$ |

| A | B | C | F |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |




(b) $F(P,Q,R) = \overline{P}\,\overline{R} + Q\overline{R} + \overline{P}Q$

→ (i) using 2:1 MUX & NOT gate :

$\overline{P}\,\overline{R} + Q\overline{R} + \overline{P}Q(R+\overline{R})$

$\overline{P}\,\overline{R} + Q\overline{R} + \overline{P}Q\overline{R} + \overline{P}QR$

$\overline{R}[\overline{P} + Q + \overline{P}Q] + \overline{P}QR$

w.k.T $A+B+AB = A+B$

$\overline{R}[\overline{P}+Q] + \overline{P}QR$

$R'(P'+Q) + R(P'Q)$

here, select line, $S=R$ in MUX-01

- $R=0 \rightarrow I_0 = P'+Q$
- $R=1 \rightarrow I_1 = P'Q$

*creating OR function for $I_0 = P'+Q$

here, select line, $S=Q$ in MUX-02

- $Q=0$ → output should be $P'$
- $Q=1$ → output should be 1

∴ output $= Q'(P') + Q(1) = P'+Q$

* creating AND function for $I_1 = P'Q$

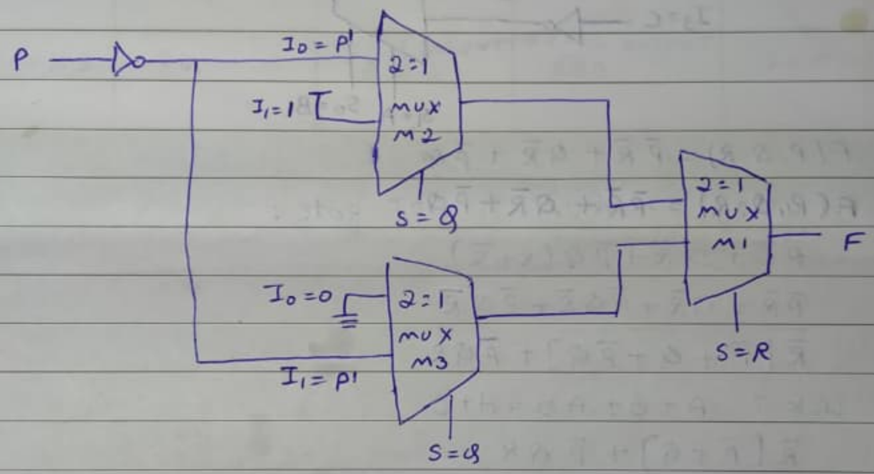
here, select line, $S=Q$ in MUX-03

- $Q=0$ → output should be 0
- $Q=1$ → output should be $P'$

∴ output $= Q'(0) + Q(P') = P'Q$

Truth Table

| P | Q | R | P' | R' | $I_0 = P' + Q$ | $I_1 = P'Q$ | $F = R'I_0 + RI_1$ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |




(ii) using 4:1 MUX:

using Select lines, $S_1 = P$ $S_0 = R$

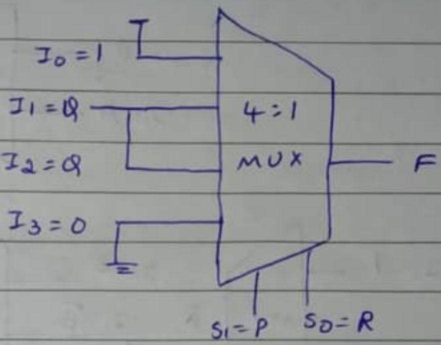
Derivation of MUX inputs

| P | R | Substitution | MUX input |
|---|---|---|---|
| 0 | 0 | $1 + Q + Q = 1$ | $I_0 = 1$ |
| 0 | 1 | $0 + 0 + Q = Q$ | $I_1 = Q$ |
| 1 | 0 | $0 + Q + 0 = Q$ | $I_2 = Q$ |
| 1 | 1 | $0 + 0 + 0 = 0$ | $I_3 = 0$ |

Truth Table:

| P | Q | R | F |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |

| P | Q | R | F |
|---|---|---|---|
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |




2) Implement the functionality of a full adder using 2:1 Muxes only.

→ The full adder has 3 inputs A, B, Cin and gives 2 outputs sum and carry.

$Sum = A \oplus B \oplus Cin$

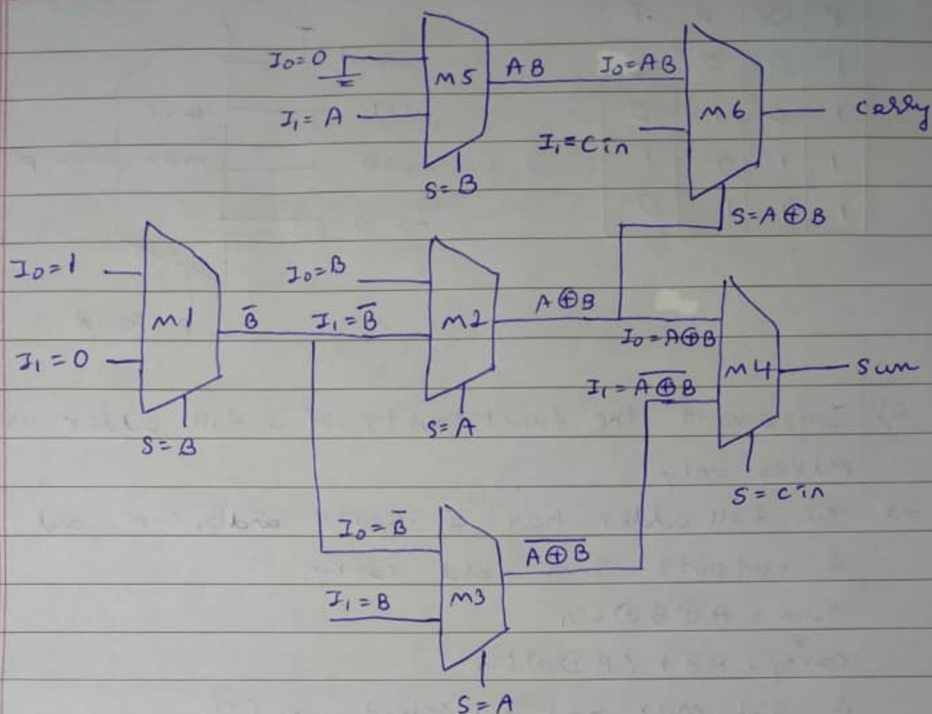
$Carry = AB + (A \oplus B)\, Cin$

A 2:1 mux has
- 1 select line (S)
- 2 inputs ($I_0$ & $I_1$)
- 1 output (Y)

$Y = \overline{S} I_0 + S I_1$

Mux connections:

| MUX | select (S) | I/P ($I_0$) | I/P ($I_1$) | output |
|---|---|---|---|---|
| MUX-1 | B | 1 | 0 | $\overline{B}$ |
| MUX-2 | A | B | MUX-1 o/p $\overline{B}$ | $A \oplus B$ |
| MUX-3 | A | MUX-1 o/p $\overline{B}$ | B | $(\overline{A \oplus B})$ |
| MUX-4 | Cin | MUX-2 o/p $A \oplus B$ | MUX-3 o/p $(\overline{A \oplus B})$ | sum |
| MUX-5 | B | 0 | A | AB |
| MUX-6 | MUX-2 o/p $A \oplus B$ | MUX-5 o/p AB | Cin | carry |




3) simplify $f(a,b,c,d) = \Sigma m(2,4,5,6,10) + D(12,13,14,15)$ and implement the SOP expression using 4:1 MUX

→ $f(a,b,c,d) = \Sigma m(2,4,5,6,10) + D(12,13,14,15)$

by using K-Map,

| AB \ CD | $\overline{C}\overline{D}$ 00 | $\overline{C}D$ 01 | $CD$ 11 | $C\overline{D}$ 10 |
|---|---|---|---|---|
| 00 $\overline{A}\overline{B}$ | 0 | 1 | 3 | 1 (2) |
| 01 $\overline{A}B$ | 1 (4) | 1 (5) | 7 | 1 (6) |
| 11 $AB$ | X (12) | X (13) | X (15) | X (14) |
| 10 $A\overline{B}$ | 8 | 9 | 11 | 1 (10) |

hence, the simplified SOP Expression is:

$\therefore \; F = B\overline{C} + C\overline{D}$

In a 4:1 MUX, there are 4 inputs $I_0, I_1, I_2, I_3$ and 2 select lines $S_0$ & $S_1$ and 1 output.

- $S_1 = C$
- $S_0 = D$

| C | D | F | MUX input |
|---|---|---|---|
| 0 | 0 | $B(1) + 0(1) = B$ | $I_0 = B$ |
| 0 | 1 | $B(1) + 0(0) = B$ | $I_1 = B$ |
| 1 | 0 | $B(0) + 1(1) = 1$ | $I_2 = 1$ |
| 1 | 1 | $B(0) + 1(0) = 0$ | $I_3 = 0$ |